

Composite coatings based on silicone-modified polyurethane and marine natural product chlorogentisyl alcohol for marine antifouling

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ARTICLE INFO

Keywords:

Composite coatings
Marine antifouling
Silicone-modified polyurethane
Chlorogentisyl alcohol
3-Chloro-2,5-dihydroxybenzyl alcohol

ABSTRACT

Marine biofouling has a serious impact on marine facilities, and traditional antifoulants may cause secondary pollution to the ocean. As chlorogentisyl alcohol (3-chloro-2,5-dihydroxybenzyl alcohol, CHBA) is a marine natural product with excellent antifouling properties, CHBA and its analogues are prepared and combined with silicone-modified polyurethane (SiPU) for development of potential green composite coatings for marine antifouling. Lubricant polydimethylsiloxane and antifoulant CHBA endow the composite material with antifouling properties under dynamic and static conditions, respectively. CHBA is steadily released from the composite coating at a rate of 28 $\mu\text{g}/(\text{cm}^2 \cdot \text{d})$. As one of the CHBA/SiPU composite coatings, SiPU-3 exhibits inspiring activities against *Escherichia coli* (Gram-negative bacteria), *Staphylococcus aureus* (Gram-positive bacteria) and *Vibrio fischeri* (marine bacteria), with all inhibition rates exceeding 90.0 %. This composite coating has high antiprotein adsorption ability and algal inhibition rate, showing desirable antifouling performance for up to 90 d in seawater. This work will provide new possible considerations in the development of marine antifouling coatings using marine natural antifoulants.

1. Introduction

Marine biofouling refers to the accumulation of bacteria, algae, and organisms on the surface of marine engineering or equipment [1]. It has detrimental effects on marine industries and facilities, and increases the operating resistance, energy consumption and carbon dioxide emissions of marine ships [2–4]. A facile and simple solution to this problem is the development of marine antifouling coatings, usually by adding antifoulants to the coatings [5–7]. With the prohibition of organotin compounds (TBT) and restrictions on antifoulants containing heavy metals copper and zinc [8–10], an urgent task is to develop effective marine antifouling coatings using environmentally friendly antifoulants, while mitigating their impact on the marine environment. Compared with heavy metal antifoulants, natural product antifoulants are often promising biodegradable, broad-spectrum, eco-friendly antifoulants [11–15], having the advantage of inhibiting the adhesion of microorganisms without killing them [16]. As a marine natural product, chlorogentisyl alcohol (3-chloro-2,5-dihydroxybenzyl alcohol, CHBA) is an isolated

from the marine *Ampelomyces* sp. strain UST040128 [17,18]. Inspiringly, CHBA has been proved to have a significant inhibitory effect on the settlement of polychaeta larvae and barnacles, with EC_{50} (concentration for 50 % of maximal effect) ranging from 0.67 to 0.78 $\mu\text{g} \cdot \text{mL}^{-1}$ and 3.19 to 3.81 $\mu\text{g} \cdot \text{mL}^{-1}$, respectively [19]. In addition, CHBA has a strong inhibitory effect on the growth of 13 common marine bacteria [20], and the non-toxic properties of CHBA to barnacles and polychaeta larvae highlight its ecological safety [21]. Furthermore, this marine natural product is simple and easy to be fabricated, and thereby has potential to be applied to solve the problem of static biofouling in marine environment.

On the other hand, silicone materials have attracted considerable attention in the field of marine antifouling due to their eco-friendliness, non-toxicity, hydrophobicity and low surface energy [22–27]. The low surface energy of the coating makes it difficult for marine organisms to penetrate and adhere [28]. However, their poor mechanical performance limits their application in complex marine environments. In addition, the removal of organisms on silicone coatings largely depends

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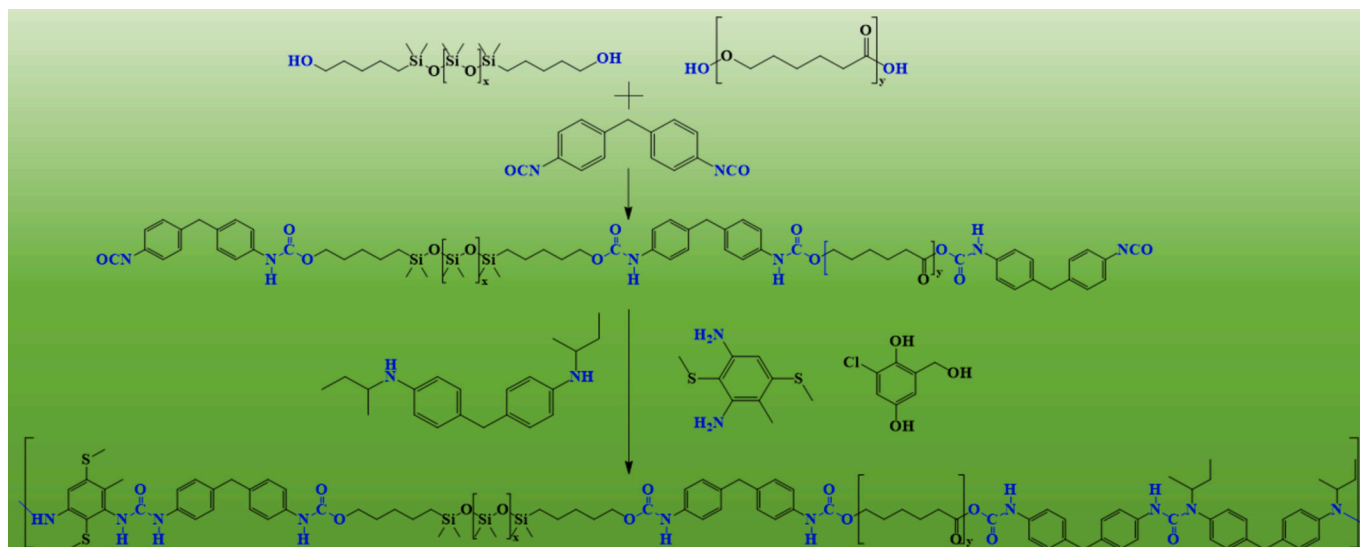
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<https://doi.org/10.1016/j.porgcoat.2024.108906>

Received 8 September 2024; Received in revised form 25 October 2024; Accepted 30 October 2024

Available online 19 November 2024

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Scheme 1. Preparation of the SiPU-X.

on the scouring and shear forces of seawater. Therefore, it shows low performance in static environment antifouling [29]. To overcome this shortcoming, the usual solution is to introduce polyurethane, urea groups or nanomaterials [30–31] into the silicone resin to improve mechanical strength and adhesion to the substrate. Polyurethane (PU) have the characteristics of high elasticity, high strength, high wear resistance, and high impact resistance [32,33]. The unique structural segment properties can limit the migration of polymer chains in water to reduce biological contamination, and PU can employ various defense strategies employed by other coatings. Webster et al. [34] developed a self-stratified cross-linked polydimethylsiloxane (PDMS)-PU coating, in which low surface energy organic silicon segment migrates to the surface during film formation, and PU component settles to the sublayer. Zhang et al. [35] prepared a PDMS-based marine antifouling PU coating, which has controllable coumarin release and UV-healing properties. Tian et al. [36] synthesized hydrophobic polyurethane coatings with the use of PDMS-OH and polyurethane, and found that the polymer not only has good anti-fouling performance, but also has anti-cavitation and anti-corrosion properties. The combination of PDMS chain segment and PU improves the coating performance while maintaining the low surface energy characteristics of PDMS. Accordingly, the combination of PU and PDMS is expected to design functional coatings with excellent antifouling ability and good adhesion.

In this context, we report a silicone-modified polyurethane antifouling coating based on marine natural antifoulant CHBA. CHBA and its representative analogues are synthesized from readily available starting materials and used for static marine antifouling through controlled release. Based on the internal microstructure of polyurethane (PU), low surface energy silicone-modified polyurethane (SiPU) was achieved via the input of silico-oxygen bonds. SiPU was fabricated using polycaprolactone diol (PCL) and hydroxy-terminated polydimethylsiloxane (PDMS-OH) as starting materials, which was used for marine dynamic antifouling. A potential eco-friendly marine antifouling composite coating has been developed by combining the antifoulant CHBA with SiPU. CHBA was enwrapped and dispersed in SiPU to test the antifouling performance of the resultant SiPU-X coatings. Among the SiPU-X coatings, SiPU-3 not only displays excellent antiprotein, antibacterial and antifungal properties, but also exhibits desirable antifouling performance in seawater for up to 90 d. Therefore, based on marine natural antifoulant CHBA and SiPU, an integrated marine antifouling composite coating with excellent antifouling performance was developed, and the formation mechanism of its microstructure was analyzed.

2. Experimental

2.1. Materials

2-Hydroxybenzyl alcohol (98 %), 2,4-dichlorophenol (99 %), 2,5-dihydroxybenzoate methyl ester (97 %), N-chlorosuccinimide (NCS, 98.5 %), lithium aluminum hydride (LAH, 97 %), and paraformaldehyde $[\text{HO}(\text{CH}_2\text{O})\text{H}]$ (95 %) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Tetrahydrofuran (THF, 99.8 %) and *N,N*-dimethylformamide (DMF, 99 %) were purchased from Anhui Zesheng Technology Co., Ltd. Dichloromethane (DCM, 95 %), ethyl acetate (EA, 96 %) and petroleum ether (PE, 95 %) were supplied by Weihai Senke Chemical Glass Instrument Co., Ltd. *N,N*-di-*sec*-butyl-4,4'-methylene dianiline (MDBA, 96 %) and 4,4'-methylene diphenyl diisocyanate (MDI, purity 99.6 %) were produced by China Wanhua Chemical Group Co., Ltd. Polycaprolactone diol (PCL, $M_n = 1000$ g/mol, 99 %) was supplied by Xuzhou Yihuiyang New Materials (China) Co., Ltd. Hydroxy-terminated butyl dimethylsiloxane (PDMS-OH, $M_n = 1750$ g/mol, 99 %) were supplied by Guangzhou Silok Polymer Co., Ltd. PCL and PDMS-OH were all dehydrated under a vacuum at 120 °C for 2 h before use. Dimethylthiololuene diamine (DMTDA, 99.5 %) was purchased from Shanghai RON Chemical Technology Co., Ltd. Artificial seawater was prepared according to ASTM D1141.

2.2. Synthesis of compounds 3 and 4

General experimental procedures. All the reactions have been done in an air atmosphere unless indicated. The solvent is distilled before use in nitrogen atmosphere. Silica gel (200–300 mesh) for flash chromatography. ^1H and ^{13}C NMR spectra were recorded on the ADVANCE III 400 MHz NMR spectrometer of the German Bruker company with CDCl_3 or CD_3OD solvent as solvent. Chemical changes are reported in ppm relative to CDCl_3 ($\delta = 7.26$ ppm) and CD_3OD ($\delta = 4.8$ ppm) for ^1H NMR and relative to the central CDCl_3 resonance ($\delta = 77.0$ ppm) and CD_3OD ($\delta = 49.0$ ppm) for ^{13}C NMR. Detailed synthesis processes and data are described in Supporting information.

2.3. Preparation of SiPU-X

Polyurethane prepolymer (PUP) was synthesized using a stepwise polymerization method [37]. To a three-neck flask with a stirrer, nitrogen device, and thermometer were added 6.0 g of PDMS-OH and 6.0 g of PCL. The resultant mixture was dehydrated under vacuum at 120 °C

for 2 h. When the reaction system was cooled to 60 °C, 4.87 g of MDI was added and stirred until evenly mixed. Then, the reaction was stirred at 80 °C for 2 h. The mass fraction of -NCO (NCO%) was titrated by di-*n*-butylamine titration. Secondly, the control chain expansion coefficient is 1.05, and the molar ratio of DMTDA to MDBA is 1:1. 0.73 g of DMTDA, 1.06 g of MDBA and 0.56 g of CHBA were uniformly mixed with 2 mL of DMF to enhance dissolution. The resultant mixture was added with PUP, stirred at room temperature for 5 min, and poured into a Teflon mold in a vacuum oven under negative pressure at 80 °C to remove bubbles. After 24 h of ripening in drying oven, the sample was obtained for testing. This sample was named SiPU-3, whose molecular structure and synthesis method are shown in Scheme 1.

In addition, samples of SiPU without the addition of CHBA, PU without the addition of PDMS-OH and CHBA, and PUP without the addition of CHBA were also prepared. Their specific synthesis process and molecular structure are shown in Fig. S2 (see Supporting information) and Table 2.

2.4. Antibacterial assay

Detailed antibacterial assay methods are described in Supporting information.

2.5. Characterizations of the SiPU-X

2.5.1. Fourier transform infrared spectroscopy (FT-IR)

Attenuated total reflection FT-IR was collected by the FT-IR spectrometer (Nicolet 380) of the American Thermoelectric Company. The sample was scanned 32 times with an average resolution of 4 cm⁻¹.

2.5.2. Mechanical experiment

The mechanical properties were tested on an electronic tensioner (Model Z005, Zwick, German) at a tensile speed of 500 mm/min. The test standard is GB/T 528-2009. The thickness of the sample was 1 mm.

2.5.3. Adhesion experiment

The adhesion strength of the coatings was analyzed using ASTM D3359 (Cross-cut tape test). According to this standard single or several cuts have been applied to the substrate by razor blades having a cutting-edge angle between 15° and 30°. The pressure-sensitive tape was located over the area of the grids and was rubbed using the pressure application device. Then, the tape was peeled off, and the grid area was inspected using an illuminated magnifier (Table S1, Supporting information).

2.5.4. Static water contact experiment

The static water contact angle measurement (WCA) was performed using a JC2000D2 goniometer (Shanghai Zhongchen Instrument Co., Ltd., China). The digital image of the droplet was obtained by the static droplet method. The result of each WCA was the median of three measurements. The surface energy (SFE) was calculated using a contact angle measuring instrument. The two liquids used were deionized water and CH₂I₂.

2.6. Antifouling experiments of the SiPU-X

2.6.1. Static protein adsorption performance test

The coating was cured on a glass slide (10 × 10 × 0.5 mm) and used for testing after 24 h of ripening. The surface of the coating was rinsed with phosphate buffer solution to remove active groups and organic contaminants. The coatings were immersed in 15 mL of 1 g/L BSA solution for static adsorption at room temperature for 24 h. After adsorption equilibrium, 100 μL of the adsorbed residual liquid was reacted with 3 mL of G-250 solution for 5 min to test the absorbance at 595 nm by UV-Visible Spectrophotometer (Lambda3665, PerkinElmer, made in American). According to the standard curve (Fig. S3, Supporting information), the adsorption amount of BSA per unit coating area

was calculated according to the concentration of BSA solution. The calculation method is as follows:

$$Q = \frac{(C_0 - C)V}{S} \times 1000$$

where Q is the adsorption capacity of the coating (μg/cm²), C₀ is the concentration of BSA solution before adsorption (1 mg/mL), C is the concentration of BSA solution after adsorption (mg/mL), V is the volume of BSA protein solution (15 mL) and S is the area of the membrane (2.2 cm²).

2.6.2. Antibacterial test

The antibacterial properties of the SiPU-X were characterized by the plate coating method. *Escherichia coli* and *Staphylococcus aureus* were used as gram-negative bacteria and positive bacteria, and in particular, *Vibrio fischeri* was used as marine bacteria to test the antibacterial properties of the coatings. These strains are frequently used as model biofouling organisms due to their broad environmental spread and for their ability for surface colonization. First, the coatings were placed into 30 mL of bacterial solution with a concentration of approximately 10⁷ CFU/mL and cultured in an incubator at 35 °C for 18 h. Second, the coatings were rinsed with a sterile PBS solution. Third, the coatings were placed in 5 mL of sterile phosphate buffer solution and sonicated for 5 min. Fourth, the obtained bacterial liquid was diluted 50 times. Then, 20 μL of each bacterial solution was poured into LB solid medium. The bacteria on LB solid medium were cultured in a constant temperature incubator at 35 °C for 18 h. Finally, colony counting was performed on the plate. In this experiment, PU was used as a control group.

2.6.3. The algal-resistance test

The diatom series *Phaeodactylum tricornutum* was selected as the experimental algae. *Phaeodactylum tricornutum* was cultured in the Si-rich sterile Guillard's F/2 medium at 26 °C and in a 12 h:12 h light and dark (L/D) cycle. The prepared coating on the cover glass was disinfected under a UV lamp for 2 h before testing. The coatings (20 × 20 × 0.5 mm) were cultured in 10⁶ cell/mL algal suspension for 7 d. Then, washed with sterile seawater to remove any unattached or poorly attached algal cells. The algal cells attached to the sample were immobilized by immersion of 2.5 % glutaraldehyde solution at 4 °C for 5 h. After fixation, the algal cells were stained with 0.05 g/L propyl iodide for 30 min, and eluted with deionized water. Finally, the coatings were photographed with an inverted fluorescence microscope (Model DMi8 Manual, Leica Microsystems, German). When counting, three fields of view were randomly selected for each sample. Three parallel groups were tested for each sample, and the average was taken.

2.6.4. CHBA release rate test

The standard curve of absorbance with CHBA concentration is shown in the S7. The characteristic UV absorption peak of CHBA was observed at 291 nm. The standard curve was constructed with the absorption peak intensity at 291 nm as the ordinate and different concentrations of CHBA as the absciss (Fig. S4, Supporting information). This method is convenient for the determination and calculation of CHBA concentration in the experiment after subsequent release. The picture shows the release curve of antifoulant embedded with CHBA, which can confirm that CHBA can achieve long-term release. From 0 to 4500 min, the antifoulant is in a growing molding process, and after 4500 min, it reaches a stable period. The concentration of CHBA was determined by ultraviolet spectrophotometry and its release rate was determined in a static environment. Dip the matured SiPU-3 coating (20 × 20 × 0.5 mm) into 200 mL ASW and let it sit for 24 h. It is worth noting that the seawater should be changed every day, and 5 mL of the solution should be taken at 1 d, 2 d, 3 d, 5 d, 7 d, 9 d, 11 d, 13 d, 15 d, 17 d, 20 d, 25 d and 30 d to determine its absorbance. The specific quantity is calculated from the established standard curve based on the peak area drawn by the

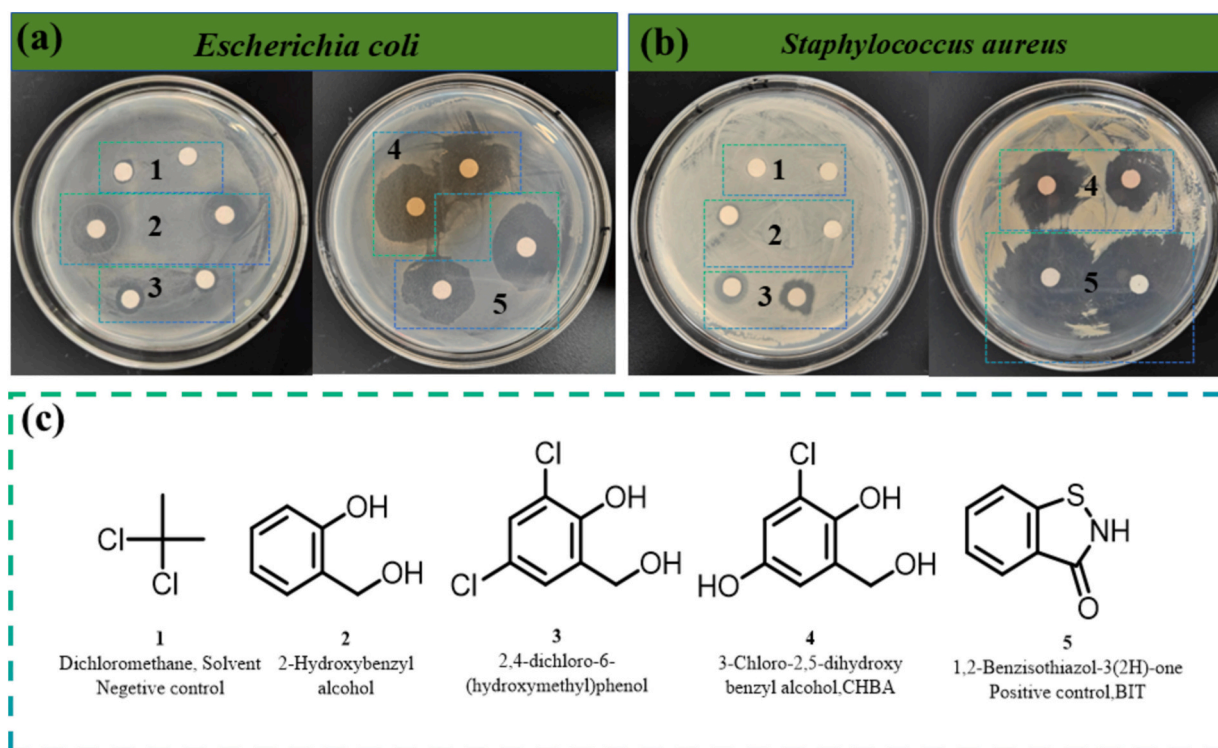


Fig. 1. Determination of antibacterial activity of 5 compounds by disk diffusion method: (a) Photo of disk diffusion experiment on *Escherichia coli*, (b) Photo of disk diffusion experiment on *Staphylococcus aureus*, (c) Structural formula of five compounds, in which compounds 1 and 5 are negative control and positive control.

Table 1

Summary of the antibacterial activity of five compounds against the bacterial species determined by disk diffusion.

Sample	1/mm	2/mm	3/mm	4/mm	5/mm
<i>Escherichia coli</i>	6.0 ± 0.0	15.0 ± 0.1	8.2 ± 0.1	20.0 ± 0.2	22.1 ± 0.1
Suppression zone	0	9.0 ± 0.1	2.2 ± 0.1	14.0 ± 0.2	16.1 ± 0.1
<i>Staphylococcus aureus</i>	6.0 ± 0.0	6.0 ± 0.0	10.2 ± 0.1	18.4 ± 0.3	30.2 ± 0.2
Suppression zone	0	0	4.2 ± 0.1	12.4 ± 0.3	24.2 ± 0.2

known standard quantity.

$$W = \frac{C \times V}{A \times D}$$

where W is the release rate of CHBA ($\mu\text{g}/(\text{cm}^2 \cdot \text{d})$), C is the content of CHBA in aqueous solution ($\mu\text{g}/\text{mL}$), V is the volume of immersion solution (200 mL), A is the surface area of coating (8.4 cm^2) and D is the immersion time (d).

2.6.5. Seawater immersion test

Reinforced epoxy resin plate ($100 \times 100 \times 2 \text{ mm}$) at 30°C and 46 % relative humidity coated sample. The thickness of the coating is controlled by the weight of the polymer solution, which is controlled at about 0.5 mm. The test panels are placed in seawater 3 m deep in the Xiaoshidao bay, Weihai, China ($37^\circ 31' 38'' \text{ N}$, $122^\circ 4' 44'' \text{ E}$). The hanging board was selected from April to June, when the Marine environment was characterized by vigorous growth and reproduction of fouling organisms, and a large number of algae and small barnacles, and the obtained solid sea hanging board data were representative. The plates are periodically removed from the sea to observe and photograph the deposition of marine life and seaweed on the plates. The epoxy resin

plate coated with PU and the board coated with SiPU are used as the control and reference respectively.

3. Results and discussion

3.1. Bacteriostatic screening of antifoulants

The antibacterial activities of dichloromethane (solvent, 1), 2-hydroxybenzyl alcohol (CHBA analogue 2), 2,4-dichloro-6-hydroxymethylphenol (CHBA analogue 3), 3-chloro-2,5-dihydroxybenzyl alcohol (CHBA, 4) and 1,2-benzisothiazol-3-ketone (positive control, 5) against *Escherichia coli* and *Staphylococcus aureus* were evaluated by disk diffusion method (Fig. 1a and Table 1). A clear inhibitory zone was observed around the disk of positive control 5, indicating a strong antibacterial activity. In contrast, under the same conditions, solvent 1 showed significant bacterial growth around the disk (negative control). Fig. 1a shows that CHBA analogue 2 only has a certain inhibitory effect on *Escherichia coli*, while CHBA analogue 3 and 4 (CHBA) have inhibitory effect on the two bacteria. Moreover, the latter has a more significant inhibitory effect than the former. In terms of molecular structure, CHBA and its analogues 2 and 3 contain the electron-donating groups of phenolic hydroxyls and hydroxymethyl, and 3 and 4 have electron-withdrawing chlorine atoms, which results in an electron cloud density order of $4 > 2 > 3$.

From their structures that have inhibitory effects on *Escherichia coli* and *Staphylococcus aureus*, the mechanism by which they inhibit bacterial growth may be that the phenoxide anion is dissociated from the phenolic group and forms a chemical bond with the active site of the bacterial membrane proteins. This can cause protein denaturation or precipitation, or inactivate the enzyme system, thereby destroying the structure and function of the membrane and reducing its viscosity.

3.2. Performance testing of coatings

Scheme 1 shows the preparation of SiPU-3. Three kinds of

Table 2

The specific composition inside the synthesized samples.

Sample	PCL/g	PDMS-OH/g	MDI/g	DMTDA/g	MDBA/g	CHBA/g
PU	6.00	/	2.81	0.63	0.91	/
SiPU	6.00	6.00	4.87	0.73	1.06	/
SiPU-1	6.00	6.00	4.87	0.73	1.06	0.18
SiPU-3	6.00	6.00	4.87	0.73	1.06	0.56
SiPU-5	6.00	6.00	4.87	0.73	1.06	0.93

polyurethane samples were prepared, namely pure polyurethane PU without antifouling system, SiPU with silicone antifouling system, and SiPU-X with silicone/antifoulant CHBA integrated system. The specific composition inside the synthesized samples is shown in Table 2.

3.3. Composite coating

The structure of SiPU was verified by comparing its FT-IR spectrum with that of the original PU polymer (Fig. 3a). The characteristic peaks of hydrogen bond N—H are observed at 3300 cm^{-1} , and the characteristic peaks of ethyl carbamate carbonyl are observed at 1727 cm^{-1} . No isocyanate peaks are found in the range of $2270\text{--}2260\text{ cm}^{-1}$, suggesting that all -NCO groups were consumed during the reaction process. The characteristic peaks of the siloxane groups at 1259 cm^{-1} (Si-CH₃ bending), 1017 cm^{-1} (Si—O stretching), and 801 cm^{-1} (Si-CH₃ wobbling) indicate that all PDMS-OH have been incorporated into the copolymer. The FTIR test results indicate that the silicone/PU copolymer coating has been successfully fabricated.

The mechanical properties of each sample were tested by tensile test

(Fig. 3b and c). The tensile strength of SiPU is higher than that of PU, but decreases with the increase of CHBA content. However, the elongation at break of all SiPU is lower than that of PU. In general, SiPU has higher tensile strength and greater strain capacity than PU. With the increase of CHBA content, the strain capacity of SiPU-X first increased and then decreased. The adhesion of the coating to the substrate is critical. The PU and SiPU were coated on epoxy resin plates with an area of $10\text{ cm} \times 10\text{ cm}$, and the substrate adhesion performance of the coatings were tested by the hundred-grid method. According to the GB/T9286-98 grid method, the adhesion performance rating table was provided in Table S2 (see Supporting information). The SiPU has a completely smooth cutting edge with obvious surface detachment, and is therefore rated as 4B (Fig. 3d). The carbamate and urea groups in polydimethylsiloxane-based polyurethane polyurea contain a large number of hydrogen bonds to form hydrogen bond networks. These hydrogen bond networks are cross-linked to greatly improve the adhesion performance of the substrate.

3.4. Antifouling performance

The antifouling strategies of SiPU-X are as follows: I) Hydrophobic and low surface energy silicon-containing surfaces; II) The existence of marine natural antifoulant CHBA. Both CHBA and SiPU have antifouling properties, but their antifouling mechanisms are different (Scheme 2). By controlling the introduction of Si—O bond to achieve the surface modification of PU, the formation of low surface energy SiPU plays a role in the dynamic anti-fouling process, which can make bacteria and algae that are difficult to adhere to organisms easier to separate. The added antifouling agent CHBA is slowly and continuously released from

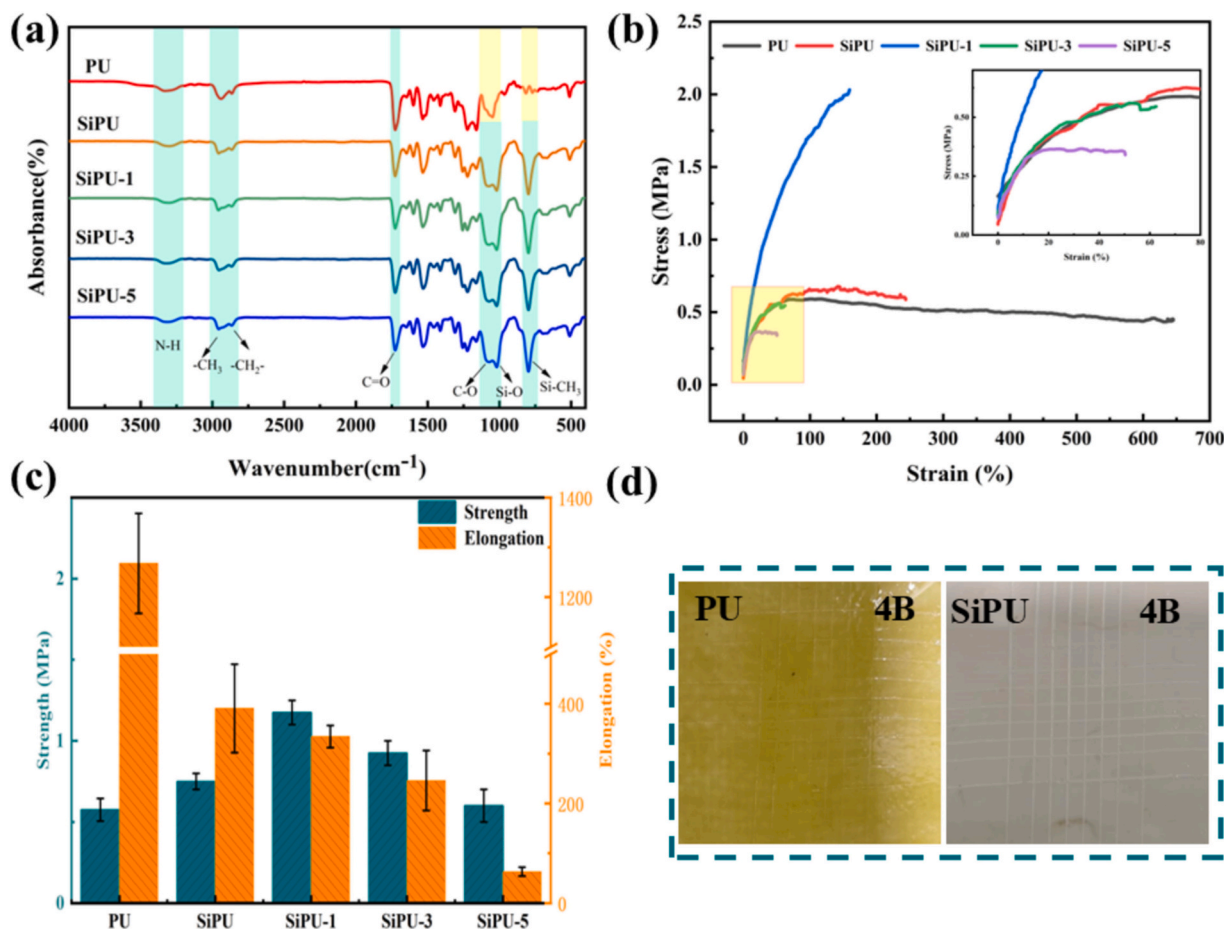
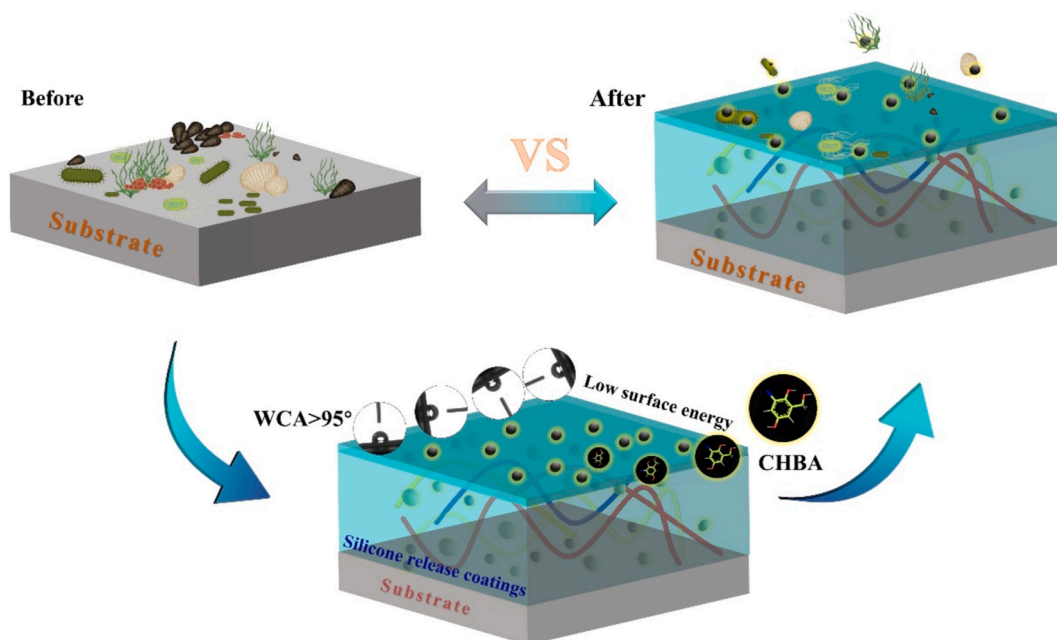


Fig. 3. (a) FT-IR spectrum of PU and SiPU-X, (b) The stress-strain curves of the samples, (c) Tensile strength and elongation at break of samples, (d) ASTM grade for adhesion of PU and SiPU tested by cross-cut tape test.



Scheme 2. Schematic diagram of the antifouling mechanism of SiPU-X.

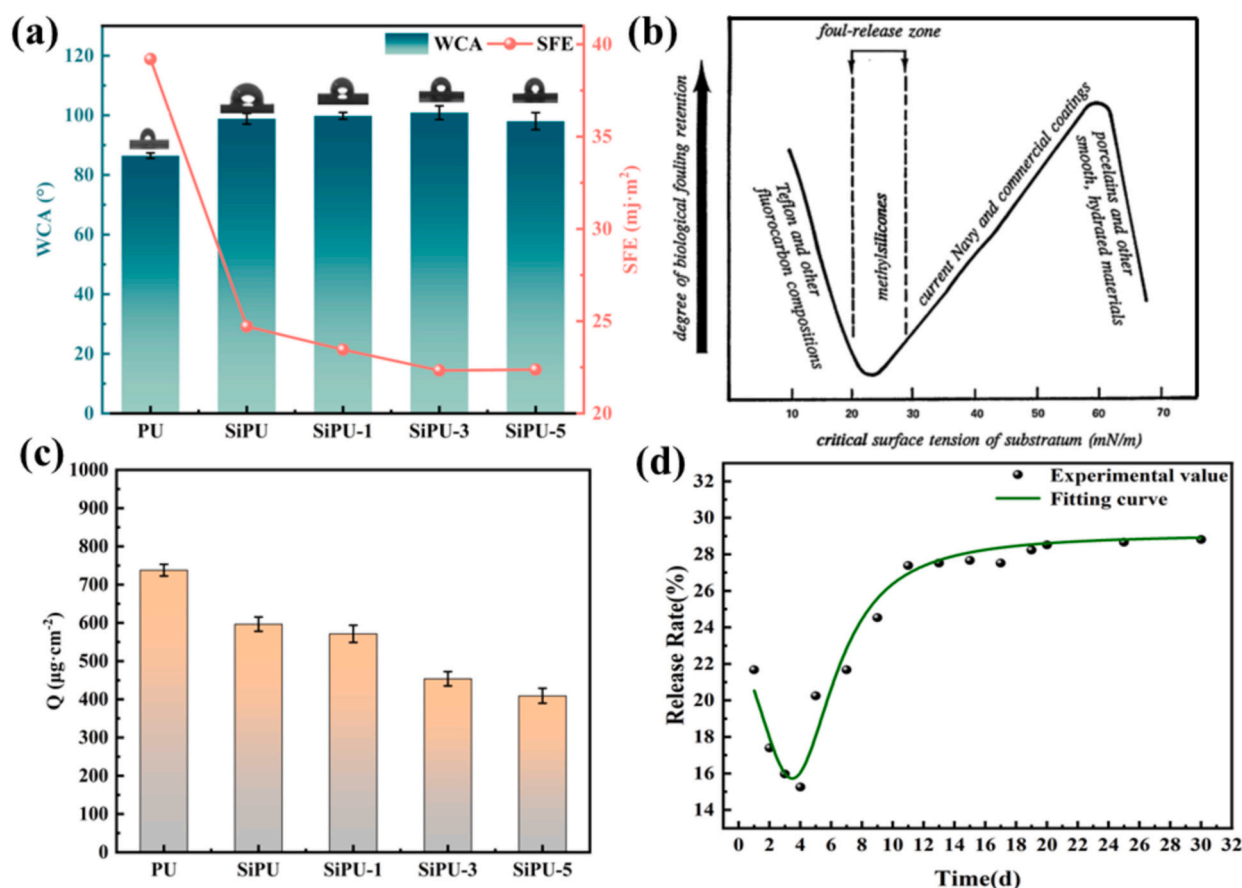


Fig. 4. (a) Water Contact Angle (WCA) and Surface Free Energy (SFE) of each sample, (b) Baier's curve of relative biological fouling as a function of surface tension of substrates [39]. (c) Static protein adsorption capacity of the coating, and (d) The release rate of CHBA at different times.

the interior into the sea water due to osmotic pressure. According to literature reports, CHBA can interact with the active site of bacterial membrane proteins to form chemical bonds [18,38], and adsorb them on microbial cells, which will lead to protein denaturation or precipitation,

or inactivate the enzyme system, thereby destroying the structure and function of the membrane and reducing its viscosity.

Fig. 4a shows the change of water contact angle (WCA) of SiPU and SiPU-1, SiPU-3, and SiPU-5 coated surfaces with different CHBA content.

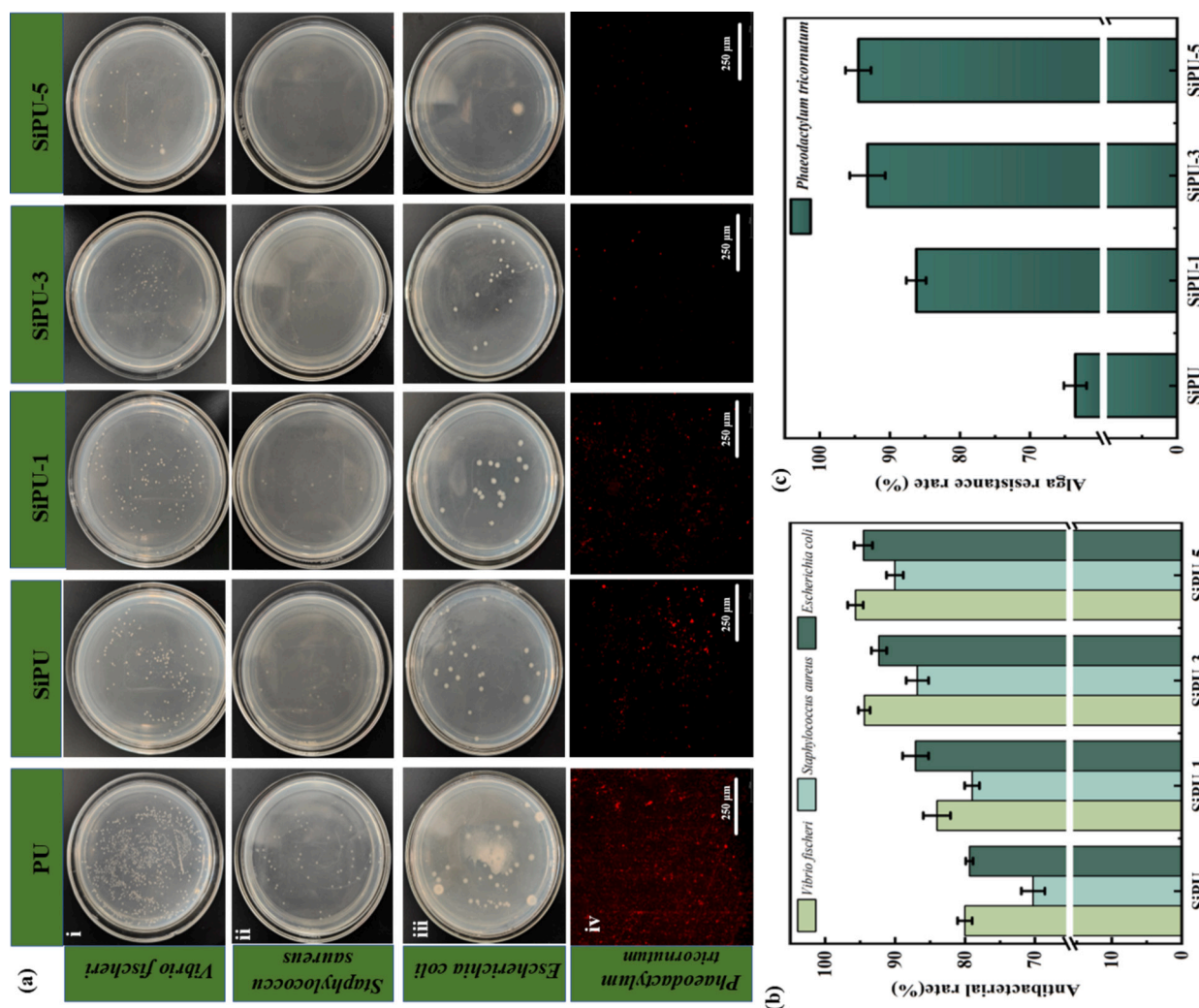


Fig. 5. Antimicrobial properties of each sample. (a) i–iii: Plate count images of the sample against *Vibrio fischeri*, *Staphylococcus aureus*, and *Escherichia coli*. iv: Digital image in a fluorescence-led micrograph of *Phaeodactylum tricornutum* attachment on the coating. (b) Antibacterial resistance rate of *Phaeodactylum tricornutum*.

It can be observed that the water contact angle of all coatings is $>95^\circ$, indicating that SiPU-X have good hydrophobic properties. There is microphase separation inside the composite coating: the silicone chain segments have lower surface energy and tend to migrate to the surface to maintain surface hydrophobicity. Instead, polyurethane segments prefer to be combined with the substrate to ensure the adhesion of the coating. With the increase of CHBA content, the angle of the coating slightly decreases, but remains stable at about 95° .

The surface energy of the coating reflects the fouling and desorption ability of the coating to a certain extent. No matter the surface energy of the coating is too large or too small, it is not conducive to the desorption of attached organisms. The biological contamination characteristics of different surfaces were studied in detail. Although its absolute level varies depending on the particular biological system, contact time, and mechanical removal force of action. By drawing the relationship between the relative amount of the biofilm and the critical surface free energy of the matrix (Fig. 4b), Bayer and colleagues [39] used the Bayer curve to reveal the trend of the influence of surface free energy on biofilm formation. They found that, at different initial critical surface tensions, the observed minimum biological adhesion strength remained between 20 and 30 mN/m ($20\text{--}30\text{ mJ/m}^2$). In this work, the SiPU surface is located in the fouling release area of the Bayer curve, indicating that the surface has a low ability to attract biological contamination. The low surface free energy weakens the interaction between the surface and biological matter, thus reducing the adhesion of biofouling [40,41]. It

difficult to wet the coating surface, affecting further contact between microorganisms and the coating, thus inhibiting the attachment of fouling organisms. It is worth noting that the surface energy of SiPU is exactly within this range. Therefore, SiPU-X is expected to have excellent adhesion resistance.

According to the mechanism of biofilm formation [1], the adhesion of proteins is first achieved through reversible adsorption of bacteria and small algae. Therefore, preventing protein from adhering to the coating surface and further colonizing bacteria is an effective antifouling strategy. With the use of PU as a control sample, Fig. 4c shows the static antiprotein adsorption capacity. Compared with PU, a small amount of protein is adsorbed on the surface of SiPU, and the addition of CHBA can also disrupt the adsorption of protein. In addition, static release test was performed on SiPU-3 (Fig. 4d). The results shows that SiPU-3 could release CHBA, and the release rate of CHBA did not change much at the initial stage of impregnation (0–4 d), then gradually increased (4–10 d), and finally stabilized (10–30 d). CHBA in SiPU-3 can be released stably, with the release rate maintained between 15 and $30\text{ }\mu\text{g}/(\text{cm}^2\cdot\text{d})$. There is a difference in the release speed between 0–4 d and 4–10 d, which may be due to the relatively high concentration of active substances in the coating and the relatively strong release driving force in the initial stage, resulting in a relatively fast release rate. As time goes by, the active substances in the coating gradually decrease, and the driving force of release weakens. At the same time, it may be affected by factors such as diffusion resistance, resulting in a lower release rate.

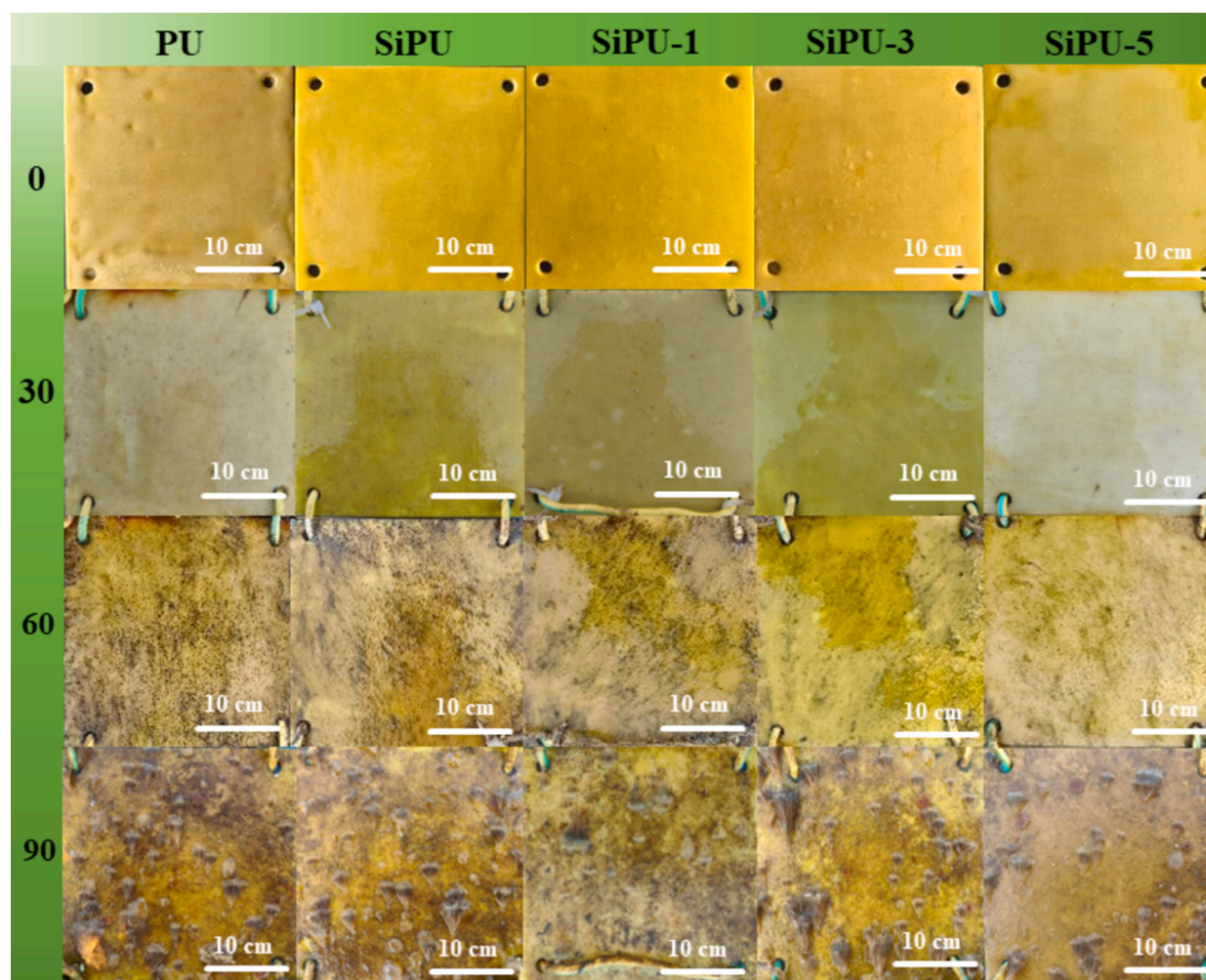


Fig. 6. Seawater immersion experiment: Typical surface antifouling images after immersion in seawater in the Xiaoshidao bay of Weihai city in China for 90 d for PU and SiPU-X.

As bacteria are important marine pollution organisms [42], *Escherichia coli* (Gram-negative bacteria), *Staphylococcus aureus* (Gram-positive bacteria), and *Vibrio fischeri* (marine bacteria, Gram-negative bacteria) are tested as representative strains to explore the antibacterial properties of the composite coating. With the number of bacteria adhering to the surface of PU as the control sample, the antibacterial performance of the coating can be reflected by the number of bacteria colonies on the plate (Fig. 5a (i–iii)). The results showed that the antibacterial properties of SiPU after antifouling treatment are improved compared with PU. The antibacterial performances of SiPU-3 and SiPU-5 are remarkable, with their antibacterial rates of *Vibrio fischeri* bacteria exceeding 90 %. The inhibition effect of SiPU-1 on *Vibrio fischeri* is not obvious since *Vibrio fischeri* itself has a certain resistance to CHBA, and the concentration used is not sufficient to achieve the inhibitory concentration. In addition, the inhibition rates of SiPU-X against *Vibrio fischeri* and *Escherichia coli* are higher than that against *Staphylococcus aureus*, indicating that CHBA has a better inhibition effect against Gram-negative bacteria.

Algae clinging to surfaces can also cause damage during marine pollution processes [43]. The inhibition effect of the composite antifouling coatings on algae was observed by fluorescence inverted microscope. Fig. 5a (iv) shows a large number and uniform distribution of *Phaeodactylum tricornutum* on the surface of PU. Compared with PU, the algal resistance of SiPU and SiPU-X is significantly improved, with the algal inhibition rates of up to 92 % (Fig. 5c). The antialgal experiments show that the continuous and stable release of CHBA and the surface

enrichment of silicone chain segments promote the removal of adhesive substances, and the mechanism is similar to that obtained by studying its antibacterial and antiprotein activities.

To test the antifouling performance of the composite coating directly, offshore field tests were carried out in the Xiaoshidao bay of Weihai city in China on May 1st, 2024. Fig. 6 shows the typical images of tested panels after immersion in the sea for 0 d, 30 d, 60 d and 90 d. The PU-coated panel was completely covered by sponges and other polluting organisms at 90 d, indicating significant contamination pressure at the testing site. In contrast, the panels coated with SiPU in the absence of CHBA showed some organism coverage at 90 d, suggesting that low surface energy modification has limited antifouling performance under static conditions. The CHBA/SiPU composite coating SiPU-3 exhibits improved antifouling performance for up to 90 d in seawater, and the use of CHBA compensates for static antifouling limitations by migrating to the surface through osmotic pressure and gradually dissolving into seawater.

4. Conclusion

In summary, marine natural antifoulant CHBA and its analogues have been designed and successfully synthesized. After screening their antibacterial activity, CHBA was chosen as the antifoulant in this work because it has the best antibacterial performance. The composite coatings were prepared by using SiPU as the substrate and dispersing antifoulant, respectively. The CHBA/SiPU composite coating SiPU-3 has

excellent antifouling performance. SiPU-3 not only has excellent anti-protein, antibacterial and anti-algal properties, but also can stabilize the release of CHBA. Indeed, CHBA can be steadily released from SiPU-3 at a rate of $28 \mu\text{g}/(\text{cm}^2 \cdot \text{d})$. The work provides a certain basis for the development of marine-natural-antifoulant-involved composite antifouling coatings. However, due to the complexity of the marine environment, there is still a long way to go for this marine antifouling coating enters the marine industry. Further optimization of CHBA-type composite coatings for marine antifouling is currently under investigation, and will be reported in due course.

CRedit authorship contribution statement

Yuan Feng: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Zi-Hao Chen:** Resources. **Yong-Yin Cui:** Writing – review & editing, Methodology. **Ming-Yang Zhang:** Data curation. **Miao-Qing Sheng:** Methodology. **Hui-Jing Li:** Resources, Funding acquisition. **Yan-Chao Wu:** Writing – review & editing, Resources, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by Harbin Institute of Technology “Double First Class” Discipline Construction Fund (2023SYLHY07), and the National Natural Science Foundation of China (22271070).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2024.108906>.

Data availability

The data underlying this study are available in the published article and Supporting information.

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